

2 – Perfect Plasticity using Hill’s anisotropic Yield Criterion

Examination guidelines

The exam for the course ‘Introduction to Theory of Materials’ consists of three parts:

- A programming task is assigned to each student. The participants are expected to implement the models assigned to them, create meaningful and representative results, and verify their implementation and results.
- The examination comprises a presentation (≈ 10 minutes).
- Oral examinations (≈ 20 minutes per person) are conducted separately for each participant. The Oral exams will contain questions to the lecture, the exercise as well as the programming task.

To give the other students insight into every examination problem, the presentations are usually given in a seminar-like manner and attended by the other students as well. However, this is not mandatory, and students can opt out of this arrangement by notifying the examiners at least one week in advance. Please be prepared to show your student ID so we can identify participants.

An exemplary workflow for the assignment is presented below:

1. Implementation of the assigned problem in *python* (other programming languages are possible, but require consultation)
2. Discussion concerning meaningful numerical examples (in addition to the results required by the assignment, if applicable)
3. Creation of representative results (e.g. plots for exemplary problems) to showcase model behaviour
4. Preparation of a presentation, including slides
5. Review of the theoretical aspects of the project with attention to the lecture content

The following deadlines hold:

- The examination will take place on September 3rd.
- The program source code has to be submitted on August 28th, 9:00 am *at the latest*, to tim.furlan@tu-dortmund.de.
- The presentation slides (.pdf, .ppt/x, .odp) have to be submitted at the day before the examination, 11.59 pm *at the latest*, to tim.furlan@tu-dortmund.de. For the presentation, the participants can either bring their own laptop or use one provided at the institute.

While we encourage you to ask questions that may arise during the assignment, please note that

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in the grading process the ability to work independently and on your own responsibility is taken into account. This particularly applies to the application of the supervisors’ recommendations and feedback to your questions. Furthermore, the code handed in is expected to be well structured (make use of subroutines), well documented (comments), and easily comprehensible (meaningful naming of functions and variables).

A typical presentation includes the following sections:

- Title (course name, name, date, ...)
- Motivation of the problem
- Relation between the assigned problem and the theory presented in the lecture, including specific model description
- Algorithmic implementation
- Representative results, including verification of the implementation (if applicable)
- Conclusion

Some guidelines for suitable presentations:

Do	Don't
• Only show core equations • Use flowcharts to represent program structure • Axes labels etc. must be readable (font size) • Use proper citations for sources • Use appropriate tools to display mathematical formulae (e.g. LaTeX beamerclass, IguanaTeX) • Use an appropriate layout for each slide	• Extensive mathematical conversions • Line-by-line presentation of program code • Use low resolution images • Forget to credit images • Use multiple text fonts and colours without a clear concept • Show nearly empty or overfull slides

Introduction and Theory

For numerous practical cases, the use of a yield function with isotropic and kinematic hardening is sufficient. Known from the lectures [1] the von Mises type yield function then takes the form

$$\Phi = \|\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}\| - \sqrt{\frac{2}{3}} [\sigma_y - \kappa] \leq 0 \quad \text{mit} \quad \boldsymbol{\sigma}_{\text{dev}}^{\text{red}} = \boldsymbol{\sigma}_{\text{dev}} + \boldsymbol{\alpha}, \quad (1)$$

and is fully isotropic, i.e. independent from the direction of (plastic) strain loading. Since, in some fields of application, the use of an isotropic yield function cannot capture the physical behaviour of certain materials anymore, one introduces a more general anisotropic format of the flow criterion

$$\Phi = \boldsymbol{\sigma}_{\text{dev}} : \mathbf{G} : \boldsymbol{\sigma}_{\text{dev}} - 1 \quad (2)$$

where $\mathbf{G}$ represents a fourth-order structural tensor. In matrix notation this can be written as

$$\Phi = [\boldsymbol{\sigma}_{\text{dev}}^{\text{m}}]^{\text{t}} \mathbf{G} \boldsymbol{\sigma}_{\text{dev}}^{\text{m}} - 1 \quad (3)$$

with

$$\boldsymbol{\sigma}_{\text{dev}}^{\text{m}} = \begin{bmatrix} \sigma_{\text{dev } 11} & \sigma_{\text{dev } 22} & \sigma_{\text{dev } 33} & \sigma_{\text{dev } 12} & \sigma_{\text{dev } 23} & \sigma_{\text{dev } 13} \end{bmatrix}^{\text{t}} \quad (4)$$

and

$$\mathbf{G} = \begin{bmatrix} F+G & -F & -G & 0 & 0 & 0 \\ -F & F+H & -H & 0 & 0 & 0 \\ -G & -H & G+H & 0 & 0 & 0 \\ 0 & 0 & 0 & 2L & 0 & 0 \\ 0 & 0 & 0 & 0 & 2M & 0 \\ 0 & 0 & 0 & 0 & 0 & 2N \end{bmatrix}, \quad (5)$$

where F , G , H , L , M and N are material parameters which characterise the anisotropy (here: orthotropy) of the material under consideration. Equation (3) thus can be rewritten as

$$\begin{aligned} \Phi = & F [\sigma_{\text{dev } 11} - \sigma_{\text{dev } 22}]^2 + G [\sigma_{\text{dev } 11} - \sigma_{\text{dev } 33}]^2 + H [\sigma_{\text{dev } 22} - \sigma_{\text{dev } 33}]^2 \\ & + 2L \sigma_{\text{dev } 12}^2 + 2M \sigma_{\text{dev } 23}^2 + 2N \sigma_{\text{dev } 13}^2 - 1, \end{aligned} \quad (6)$$

which represents the so-called *Hill's orthotropic flow criterion*, cf. [2]. The material parameters F , G , H , L , M and N in this case can be conveniently determined in experiments via the measurement of *normal* flow stresses σ_y^{11} , σ_y^{22} and σ_y^{33} as well as the *shear* flow stresses σ_y^{12} , σ_y^{23} and σ_y^{13} which as a result then read

$$F = \frac{1}{2} \left[\frac{1}{[\sigma_y^{11}]^2} + \frac{1}{[\sigma_y^{22}]^2} - \frac{1}{[\sigma_y^{33}]^2} \right], \quad (7)$$

$$G = \frac{1}{2} \left[\frac{1}{[\sigma_y^{11}]^2} + \frac{1}{[\sigma_y^{33}]^2} - \frac{1}{[\sigma_y^{22}]^2} \right], \quad (8)$$

$$H = \frac{1}{2} \left[\frac{1}{[\sigma_y^{22}]^2} + \frac{1}{[\sigma_y^{33}]^2} - \frac{1}{[\sigma_y^{11}]^2} \right], \quad (9)$$

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and

$$L = \frac{1}{2[\sigma_y^{12}]^2}, \quad M = \frac{1}{2[\sigma_y^{23}]^2}, \quad N = \frac{1}{2[\sigma_y^{13}]^2}. \quad (10)$$

By setting

$$F = G = H = \frac{1}{2\sigma_y^2} \quad (11)$$

$$L = M = N = \frac{3}{2\sigma_y^2}, \quad (12)$$

the von Mises yield function (1) can be recovered as a special case of Hill’s yield criterion.

Task

1. Derive the necessary algorithmic relations for the above Hill’s yield function, assuming *perfect plasticity*, so do not consider any hardening behaviour.
2. Implement the above material model in *python* based on the constitutive driver used in the tutorials, applying load cases depicted in fig. 1 and making use of the set of material parameters shown in table 1. The exact algorithmic tangent operator shall **not** be implemented here. For simplicity, use a numerically determined tangent operator based on a forward differential as shown, e.g., in the routine `numericalTangent()` in the tutorials. Note that, in order to implement a numerical tangent, it is convenient to implement two separate functions for the calculation of stresses and the tangent operator, respectively.

To be able to investigate the anisotropic behaviour, an *equi-biaxial strain state* in $\mathbf{e}_1$ - and $\mathbf{e}_2$ -direction is to be applied, cf. fig. 1. Therefore, the strain-driven update in every time step has the format

$$\varepsilon_{n+1} := \varepsilon_{11}(t_{n+1}) = \varepsilon_{22}(t_{n+1}). \quad (13)$$

Note that, in order to apply this load case, the function `partition()` used in the stress driver routine has to be modified accordingly!

3. After successful implementation, investigate and interpret the results from the above model and plot the results in an illustrative manner, such as the total ε_{11} , ε_{22} or plastic strains ε_{11}^p , ε_{22}^p over time t , strains ε_{11} , ε_{22} or stresses σ_{11} , σ_{22} . Furthermore, check if under usage of equations (11) and (12), the von Mises flow criterion can be recovered.

Hints

1. Due to the structure of the yield function, the *explicit* calculation of the consistency parameter is no longer possible, i.e. $\Phi_{n+1} = 0$ cannot be solved explicitly with respect to $\Delta\lambda_{n+1}$. The latter is to be determined iteratively, using a Newton-Raphson iteration scheme known from the tutorials.

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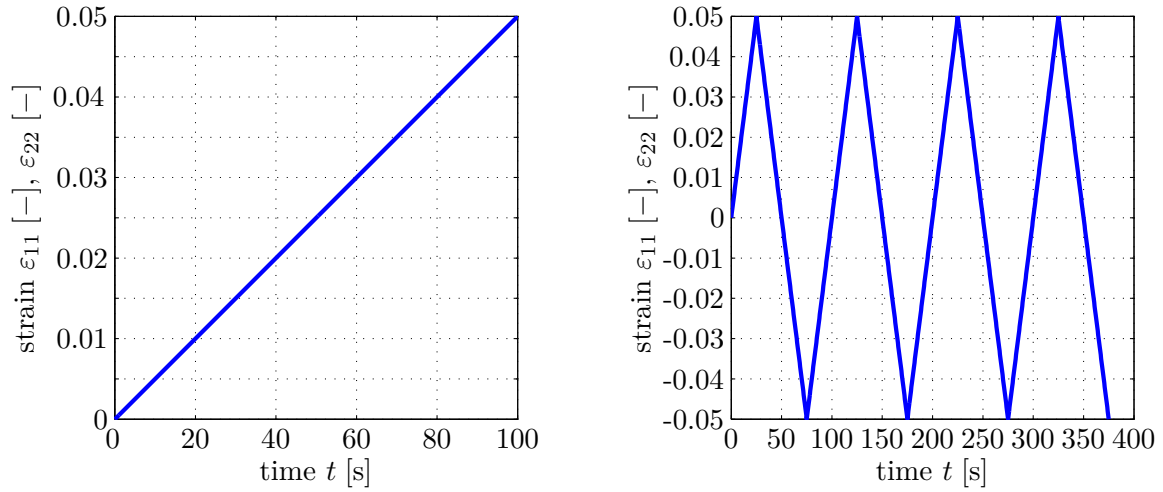


Figure 1: Equi-biaxial load cases to be investigated: Ramp function (left) and cyclic load (right).

Table 1: Material parameters to be used in the simulations.

Parameter	E	ν	σ_y^{11}	σ_y^{22}	σ_y^{33}	σ_y^{12}	σ_y^{23}	σ_y^{13}
Value	200000	0.3	580	510	460	590	530	550

References

- [1] A. Menzel. Theory of Materials. Lecture Notes, TU Dortmund, Institute of Mechanics.
- [2] R. Hill. The Mathematical Theory of Plasticity. Oxford University Press, 1950.
- [3] A. de Souza Neto, D. Perić and DRJ. Owen. Computational Methods for Plasticity: Theory and Applications. John Wiley & Sons, 2008